SLEEP-EDF · STAGING + VERIFIED REPORTS · 14 OCT 2026

Our sleep system against published work

The staging student is in the upper-middle of published Sleep-EDF-78 results at 0.18M parameters, and joint-best on N1 and REM. What sets the system apart is what comes after staging: a local report model whose every claim is verified before release.

Ⓜ measured by us Ⓟ published by the authors Ⓔ estimate or target, not yet measured

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The three systems

EXISTING SYSTEMS

Published staging models and LLM studies

Accuracy (EDF-78)	78.5–85.0%
κ	0.73–0.79
Parameters	0.049M–21M
Written report	general LLMs
Report checked first	not found
Edge	staging only

Values Ⓟ; protocols differ between papers.

OUR SYSTEM TODAY

EEG + EOG student, verified local reports

Accuracy, CV SC	83.5%
κ, CV SC	0.776
Test (29 nights)	81.96%, κ 0.756
Parameters	175,599
Report, shipped	29/29, 0 violations
Numeric fidelity	1.000

All values Ⓜ; x86 CPU, offline.

PROPOSED

Plus agentic layer and Raspberry Pi 5

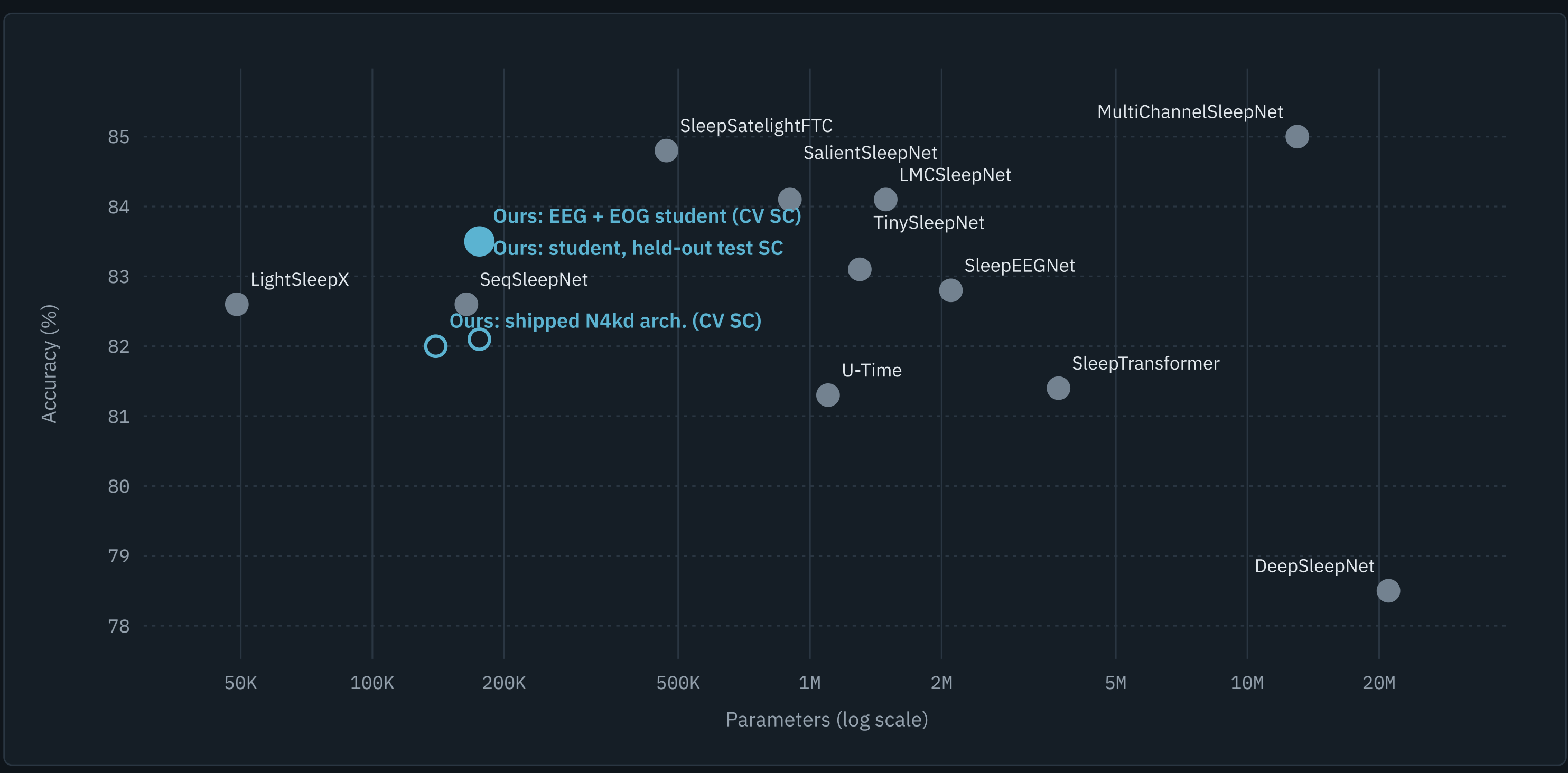
Staging	same model
EOG missing	κ 0.744 (measured)
Night on Pi 5	~6–9 min
Peak memory	~6–8 GB
Fault scenarios	15 planned
Safety invariants	8 planned

Pi values Ⓔ; not built yet.

Performance against model size

Sleep-EDF-78. Published points use 10- or 20-fold CV on SC subjects. Ours is 5-fold CV over SC + ST, shown for its SC subjects, with corrected labels, so it is close but not identical. The x-axis is logarithmic.

Accuracy | Macro-F1 | Cohen's κ



Overall metrics

MODEL	INPUTS	PARAMS	ACC %	MACRO-F1	Κ	SOURCE
DeepSleepNet	EEG	21M	78.5	75.3	0.730	Ⓟ
SleepEEGNet	EEG	2.1M	82.8	77.0	0.730	Ⓟ
SleepTransformer	EEG	3.7M	81.4	74.3	0.740	Ⓟ
U-Time	EEG	1.1M	81.3	76.3	0.750	Ⓟ
SeqSleepNet	EEG	0.164M	82.6	76.3	0.760	Ⓟ
TinySleepNet	EEG	1.3M	83.1	78.1	0.770	Ⓟ
LightSleepX	EEG + EOG	0.049M	82.6	78.7	0.770	Ⓟ
LMCSleepNet	2 EEG + EOG	1.49M	84.1	77.7	0.770	Ⓟ
SalientSleepNet	2 EEG + EOG	0.900M	84.1	79.5	–	Ⓟ
SleepSatelightFTC	EEG	0.470M	84.8	–	0.787	Ⓟ
MultiChannelSleepNet	2 EEG + EOG	13M	**85.0**	**79.6**	**0.790**	Ⓟ
Ours · shipped N4kd arch. (CV SC)	EEG	0.140M	82.0	77.6	0.756	Ⓜ
Ours · EEG + EOG student (CV SC)	**EEG + EOG**	**0.176M**	**83.5**	**79.3**	**0.776**	Ⓜ
Ours · student, held-out test SC	EEG + EOG	0.176M	82.1	77.3	0.754	Ⓜ

Green marks the best value in each column. The best published model, MultiChannelSleepNet, is +1.5 accuracy and +0.014 κ above ours, with 74× more parameters and an extra EEG channel.

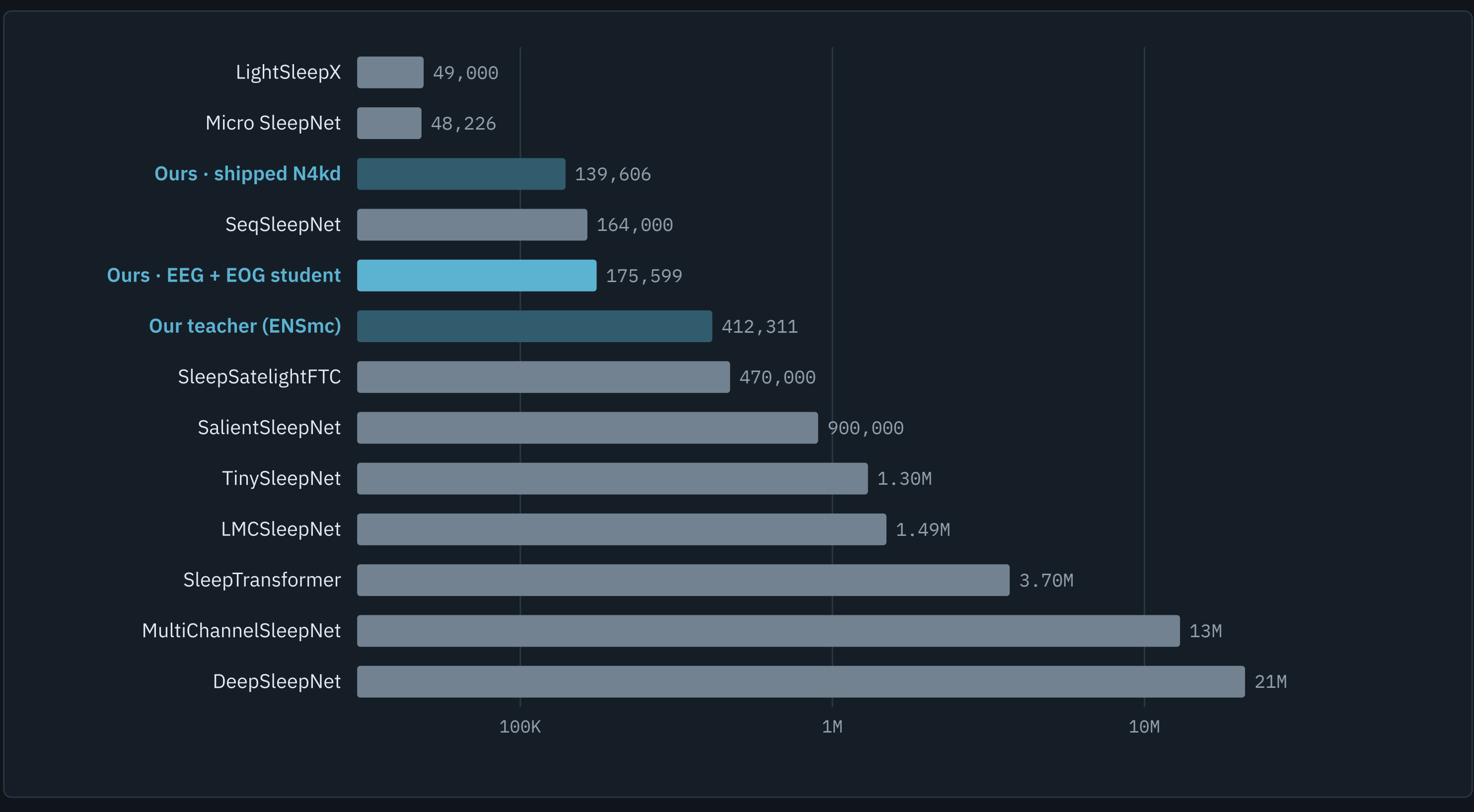
Per-class F1

MODEL	PARAMS	W	N1	N2	N3	REM
DeepSleepNet	21M	91.0	47.0	81.0	69.0	79.0
SleepEEGNet	2.1M	90.3	44.6	85.7	81.6	82.9
TinySleepNet	1.3M	92.8	51.0	85.3	81.1	80.3
SalientSleepNet	0.9M	93.3	**54.2**	85.8	78.3	**85.8**
LMCSleepNet	1.49M	**94.2**	48.5	85.8	77.6	82.7
MultiChannelSleepNet	13M	94.0	53.0	**86.9**	**81.8**	82.6
Ours · EEG + EOG student, CV SC	**0.176M**	**93.1**	**54.1**	**84.3**	**79.0**	**85.8**
Ours · student, held-out test (29 nights)	0.176M	91.4	50.7	84.3	80.9	83.7
Ours · shipped N4kd, held-out test	0.140M	90.7	45.5	81.9	79.9	78.2

Shading follows each column's range. N1 and REM, the stages EOG helps, are joint-best (54.1, 85.8). N2 and N3 are our relative weaknesses, 1–3 points below the best.

HELD-OUT TEST, CORRECTED LABELS	W	N1	N2	N3	
Shipped N4kd · P / R / F1	96.8 / 85.3 / 90.7	40.8 / 51.6 / 45.5	81.1 / 82.7 / 81.9	78.3 / 81.5 / 79.9	79.9 / 76.5
EEG + EOG student · P / R / F1	**95.9 / 87.2 / 91.4**	**47.6 / 54.3 / 50.7**	**82.7 / 86.1 / 84.3**	**79.4 / 82.4 / 80.9**	**86.2 / 81.3**
Change in F1	**+0.7**	**+5.3**	**+2.4**	**+1.0**	

Model size



Student vs teacher. Our 175,599-parameter student is better than its 412,311-parameter teacher on every test metric: 81.96% vs 80.02% accuracy, κ 0.756 vs 0.732. That's 43% of the teacher's size. The usual distillation result is a student that comes close to its teacher; the EOG branch and training recipe take this one past it.

The whole system isn't small. The staging model is 0.7 MB. The report model (Llama-3.2-3B-Student, Q4_K_M) is 2.0 GB on disk and peaks at about 4.7 GB of RAM on x86. On an edge device, the report model is the cost.

Capability by capability

CAPABILITY	EXISTING SYSTEMS	OUR SYSTEM TODAY	PROPOSED
Written report	General LLMs reading PSG reports (ChatGPT-4o; local LLMs for OSA, judged not ready) Ⓟ	Local 3B model writes evidence-cited claims under a grammar Ⓜ	Same, plus fallback summary from packet values Ⓔ
Checked before release	Not found in the work searched	Frozen verifier, 30 codes; fidelity 1.000; shipped test 29/29 Ⓜ	Same gate plus 8 agent invariants Ⓔ
Uncertainty in the output	Rare (SleepTransformer reports uncertainty) Ⓟ	Calibration, per-stage reliability, night tier, N1 flag; unreliable values must be hedged Ⓜ	Plus signal-quality reasons Ⓔ
Missing channel	Usually not reported	EOG off: κ 0.744, still above the shipped model Ⓜ	Agent routes automatically Ⓔ
Bad signal	Goes to the model	Goes to the model	QC step: review with a reason code Ⓔ
LLM runtime	Cloud or GPU	x86 CPU, about 70–115 s per report. The evaluated `llama-completion` path scores 30/31 on dev; `llama-server` scores 20–23/31 Ⓜ	Evaluated path on Pi 5, about 5–8 min per report Ⓔ
Edge device	Staging only on phones and MCUs (Micro SleepNet 48K params, 2.8 ms/epoch) Ⓟ	x86 PC	Staging + verified report on Raspberry Pi 5, about 6–9 min per night Ⓔ
Workflow	Human-run tools	Script pipeline	Agent: QC → study period → routing → staging → report → fallback → audit Ⓔ
Audit	Rarely described	Hash-pinned components, audit log, human-review routing Ⓜ	Per-step trace, policy hash, resume after power loss Ⓔ
Validation	Dataset CV	CV, held-out test, logged test access Ⓜ	Simulation ladder: Docker → ARM64 emulation → overnight replay with faults → real Pi Ⓔ

Is there novelty?

This assessment comes from targeted literature searches on 14 October 2026, not a systematic review. Any paper claim should be phrased "to our knowledge".

Established, so not claimed

Lightweight CNN + Transformer staging
Many published models, some smaller than ours.

EEG + EOG fusion
Used by LightSleepX, SalientSleepNet, LMCSleepNet and others.

Knowledge distillation and multi-scale front ends
Standard techniques.

Grammar-constrained decoding and local LLMs
General tooling.

Plausibly new

Verified, evidence-cited sleep reports from a small local LLM
Every claim cites a packet field and must pass a frozen verifier before release. Shipped test: 29/29, 0 violations, fidelity 1.000. The same base model without this training scores 2/31.

Uncertainty carried into the report and enforced
Unreliable values must be hedged, and reliable ones must not be. The verifier checks both.

A sub-200K student that beats its 412K teacher
Gated EEG + EOG fusion that starts from the distilled function and still works without EOG.

A label-alignment finding
The drop-then-trim preprocessing shifted 5–7% of labels. Fixing it alone is worth +1.3 to +3 points.

Proposed: an offline, auditable agent on a single-board computer
Edge staging exists; we found no end-to-end staging and verified reporting on a Pi-class device. It isn't built yet.

To make the claim stronger: run the standard Sleep-EDF-78 protocol once; evaluate on an external dataset; measure on a real Pi 5; and report violation rates of the same LLM with and without the grammar and verifier.

Sources

1. LMCSleepNet: Tables 4–5, per-class F1, parameters, channels
2. LightSleepX: Tables 1–2, Sleep-EDF-20/78 comparison
3. SleepSatelightFTC
4. Micro SleepNet: mobile, 48K parameters
5. ChatGPT-4o on PSG interpretation
6. Locally run LLMs for OSA
7. BrainAgent · SleepLM · L-SeqSleepNet
8. Our results: `COMPARISON WITH EXISTING SYSTEMS/COMPARISON_WITH_EXISTING_SYSTEMS.md`, `ADDING EOG VERSION - FINAL SUMMARY/numbers.json`

Published figures are taken from comparison tables in LMCSleepNet and LightSleepX, which sometimes disagree with each other (AttnSleep's size, for example). Check the original papers before citing.